Exercise 6

Donnerstag, 11. Dezember 2025 08:02

Turning Flight II

$n_V = 0$

Assume horizontal flight. Consider the following turbojet aircraft:

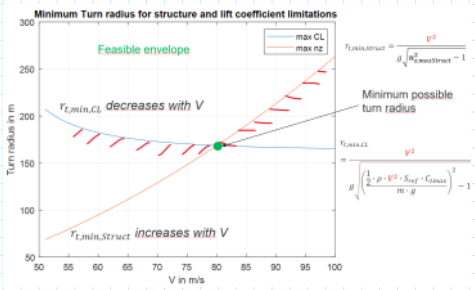
Aircraft mass	m	2000 kg
K-factor	k	0,1
Reference wing area	S_{ref}	20 m²
Zero-lift drag coefficient	C_{D0}	0,023
Maximum lift coefficient	$C_{L,max}$	1
Maximum load factor	$n_{z,max}$	4
Thrust density dependency exponent	n_p	0.75
Maximum thrust at MSL	$T_{max,S}$	0.55 · m · g

a) What is the minimum turn radius (i.e. the turn radius where $C_{L,max}$ and $n_{z,max}$ are not exceeded) and the corresponding aircraft speed? Assume a height h = 0m.

$(n_{z,max})_{C_L}$

Structural Load: $r_t = \frac{V^2}{g\sqrt{n_z^2 - 1}} \rightarrow r_{t,min,struct} = \frac{V^2}{g\sqrt{(n_{z,max})^2 - 1}}$

Aerodynamic Limit: $r_t = \frac{V^2}{g\sqrt{\left(\frac{L}{mg}\right)^2 - 1}} \rightarrow r_{t,min,C_{Lmax}} = \frac{V^2}{g\sqrt{\left(\frac{\frac{1}{2}\rho V^2 S_{ref} C_{Lmax}}{mg}\right)^2 - 1}}$



$$r_{t,min} = \frac{V^2}{g\sqrt{n_{z,max}^2 - 1}} = \frac{V^2}{g\sqrt{(n_{z,max})^2 - 1}} = \frac{V^2}{g\sqrt{\left(\frac{\frac{1}{2}\rho V^2 S_{ref} \cdot C_{Lmax}}{mg}\right)^2 - 1}}$$

At what Velocity do we achieve the maximum load factor ($n_{z,max}$) while being at aerodynamic limit (C_{Lmax})?

$n_z = \frac{L}{mg} = \frac{\frac{1}{2}\rho V^2 S_{ref} \cdot C_L}{mg}$ → C_{Lmax}

$n_{z,max}$

$\hookrightarrow V_{t,min} = \sqrt{\frac{2 \cdot n_{z,max} \cdot m \cdot g}{\rho \cdot S_{ref} \cdot C_{Lmax}}} = 80.04 \text{ m/s}$

$h = 0 \rightarrow \rho_{ISA} = 1.225 \frac{kg}{m^3}$

ρ_{ISA}

$r_{t,min} = \frac{V_{t,min}^2}{g\sqrt{n_{z,max}^2 - 1}} = 168.62 \text{ m}$

b) Can this turn radius be achieved with the available propulsive power? How high is the required thrust? Assume a height h = 0m.

$T = D$

C_{Lmax}

$T_{req} = \frac{1}{2}\rho V^2 S_{ref} C_D = \frac{1}{2}\rho_{ISA} \cdot V_{t,min}^2 \cdot S_{ref} \cdot (C_{D0} + k \cdot C_L^2) = 9652{,}85 \text{ N}$

$T_{avail} = 0.55 \cdot m \cdot g = 10791{,}00 \text{ N}$

$\rightarrow T_{req} < T_{avail} \rightarrow$ Turn is achievable

c) At what height is the available thrust not high enough anymore to provide the thrust required for the possible minimum turn radius? How high is the turn radius at this altitude? Use the standard atmosphere.

$T = D = \frac{D}{L} \cdot L = \frac{\frac{1}{2}\rho V^2 S_{ref} C_D}{\frac{1}{2}\rho V^2 S_{ref} \cdot C_L} \cdot L = \frac{C_D}{C_L} \cdot n_z \cdot mg$

→ for this case the required thrust does not depend on altitude

$$T_{req} = \frac{c_{D,max}}{c_{L,max}} \cdot n_{z,max} \cdot mg$$

$$T_{avail} = T_{ref} \cdot \underbrace{\left(\frac{V}{V_{ref}}\right)^{n_p}}_{=1} \cdot \left(\frac{\rho}{\rho_{ref}}\right)^{n_p} = T_{ref} \cdot \left(\frac{\rho}{\rho_{ref}}\right)^{n_p}$$

→ At point of interest: $T_{avail} = T_{req}$: $\frac{T_{req}}{T_{ref}} = \left(\frac{\rho_{crit}}{\rho_{ref}}\right)^{n_p}$ with ρ_{crit} representing the density at the desired altitude.

$$\rho_{crit} = \left(\frac{T_{req}}{T_{ref}}\right)^{\frac{1}{n_p}} \cdot \rho_{ref} = 1.0559 \frac{kg}{m^3};$$

$$\rho = \rho_{sea}\left(1 + \frac{\gamma_{TR}}{T_{sea}} \cdot H_G\right)^{\frac{1}{n_{TR}-1}} \rightarrow H_G = \left(\left(\frac{\rho}{\rho_{sea}}\right)^{n_{TR}-1} - 1\right) \cdot \frac{T_{sea}}{\gamma_{TR}} = 1512.2 m;$$

$$H = \frac{r_E \cdot H_G}{r_E - H_G} = \text{[illegible]};$$

$V^2 \sim 1/\rho$

→ $V_{r,min}$: $\frac{V_{sea}^2}{V^2} = \frac{\rho}{\rho_{sea}} \rightarrow V = V_{sea} \cdot \sqrt{\frac{\rho_{sea}}{\rho_{crit}}} = 86.21 m/s;$

$$\rightarrow r_{t,min} = \frac{V^2}{g\sqrt{n_z^2} - 1} = 195.63 m;$$

$n_{z,max}$